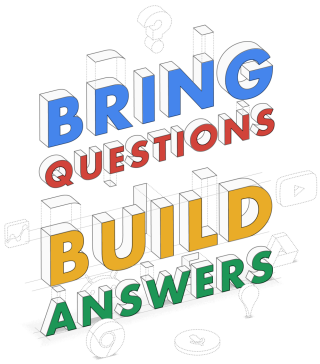


Interviewing for the Technical Consultant role

About gTech: Please watch this Youtube presentation

We partner with Sales Executives and Product Specialists and engage with agencies, clients, publishers and partners to identify creative opportunities, champion technical innovation, drive partner acquisitions, manage large, branded advertiser and marketing programs, and orchestrate technical data integrations.

Multi-product display consultancy to champion innovation - the expert in the room

- We attend client meetings and agency briefings to consult on the technical capabilities of our brand offerings and ad products, respond to digital briefs and workshop high-impact programs, adding a Google technology lens to an advertiser or agencies idea
- We provide technical and business related guidance and expertise on brand solutions to our internal partners and external customers to improve service quality for our top tier advertisers.
- We are there to push technical innovation, whilst also flagging potential challenges and articulating the technical complexity of implementations, to avoid last minute fire drills and ensure all stakeholders are well-informed
- Having us in the room allows Sales to focus on product narratives and relationship management, because we provide the technical muscle that is ultimately critical to winning the deal and winning in brand.

Management of complex implementations and technical programs

- Whilst there are multiple stakeholders involved in the execution of large programs, we are the main point-of-contact accountable for successful implementation, from planning through to post-campaign reporting, in order to grow YouTube & Google O&O revenue and market share, and get the right product to the right customers through the right partners.
- We analyze the effectiveness of programs and conduct reporting and ROI analysis to identify new creative opportunities based on sales and industry trends and customer needs
- We implement consistent best practices across product areas and enhance team to focus on higher value activities.

Evangelize process improvements and product knowledge with an eye towards scaling and effectiveness

We uplevel proactive, consultative support for high value partner segments, and identify opportunities for scaling our products and support through process improvements and refinements and data insights. We close the feedback loop for clients and Sales teams by escalating bugs and feature requests to our technical support teams.

What to expect from your technical interviews:

1) **TSC Interview(s)**

Web Technologies - Knowledge of Front End programming languages (HTML, CSS, JavaScript), protocols (HTTP and TCP/IP) and general understanding of how the internet works. Basic Unix/Linux.

- The communication protocols, languages/APIs, and other mechanisms that enable the internet to function.
- HTTP, Browsers, DNS, HTML/XML, AJAX, etc
- HTTP Protocol basics: Part I, Part II
- To study you should be able to up an HTTP server and enable basic authentication, HTTPS and cookies and take a network capture to see how these mechanisms work.

Troubleshooting/Problem Solving - How you handle certain situations with stakeholders and clients. They may give you a problem to solve (i.e something is broken, how do you go about fixing it) or ask for you to detail to them a problem you have had to solve.

2) SWE Interview:

Coding Ability - There will be a shared document between you and the interviewer, to facilitate any coding questions you may get asked. The code needs to be compilable, but you don't need any software/an IDE for this. The code will just be on this document. You can choose any language to code in (we use Java mainly, but C++ and Python are also good alternatives - other languages are also fine).

Algorithm Complexity:
Review Big-O notation. That stuff is useful every time we write an algorithm at Google, so it's great when candidates understand it!

Sorting:
You should know the details of at least one n*log(n) sorting algorithm, preferably two (say, quicksort and merge sort). Merge sort can be highly useful in situations where quicksort is impractical, so take a look at it.

Hashtables:
Familiarise yourself with this very useful data structure! Consider when and why you might use a hash table as well as how they work.

Trees:
Know about trees; basic tree construction, traversal and manipulation algorithms. Familiarise yourself with binary trees, n-ary trees, and trie-trees. Be familiar with at least one type of balanced binary tree, whether it's a red/black tree, a splay tree or an AVL tree, and know how it's implemented. Understand tree traversal algorithms: BFS and DFS, and know the difference between inorder, postorder and preorder.

Other data structures:
You should study up on as many other data structures and algorithms as possible. You should especially know about the most famous classes of NP-complete problems, such as traveling salesman and the knapsack problem, and be able to recognize them when an interviewer asks you them in disguise. Find out what NP complete means.

Systems Design:
Know Google's products, and think about how you would design the back-end (of front-end). System design questions are a test of your problem solving. Sample topics include: features sets, interfaces, class hierarchies, distributed systems, designing a system under certain constraints, simplicity and robustness, tradeoffs.

Coding:
You should know at least one programming language well, and it should preferably be Python or Java. C# is OK too, since it's pretty similar to Java. You will be expected to write code in at least some of your interviews. You will be expected to know a fair amount of detail about your favorite programming language and also be expected to produce code on a whiteboard - although you won't be expected to know all of the standard libraries off by heart!

- List of Algorithm general topics
- List of Algorithms
- List of Data Structures

Classic CS problems

Towers of Hanoi - Write a recursive program to solve this problem. What is the complexity?
Shortest path problem - Write an algorithm to plan a route by minimising distance or time.
Traveling salesman problem - Write an algorithm to determine the least cost round-trip, given multiple cities and varying costs of flights.
Knapsack problem - Write an algorithm to optimize the value of items you can fit into a backpack basedon weight and volume.

Tips

- Start preparing early. Your interview will be mainly technical, so give yourself at least 2 weeks to review and hone your coding, algorithmic and problem solving skills.
- Review your data structures and algorithms textbook, paying special attention to the topics listed above.
- Write out some algorithms yourself from start to finish and make sure they're working.
- Think about practicing some of the following programming questions in small groups….

Sample questions from *Programming Pearls*

1. Given a file containing at most ten million 7-digit integers with no duplicates. What is an efficient way to print these numbers in ascending order using just 1.5Mb RAM and reading the data just once? What are the consequences of only having 1Mb of RAM and no other storage? How would your answer change if duplicates were permitted?
2. Given a sequential file that contains at most four billion 32-bit integers in random order, find a 32-bit integer that isn't in the file (and there must be at least one missing...why?). How would you solve this with unlimited main memory? How would you solve it if you could use several external files but only a few bytes of main memory?
3. Rotate a one-dimensional vector of n elements left by i positions. For instance, with n=8 andi=3, the vector abcdefg is rotated to defghabc. Simple code uses an n-element intermediate vector to do the job in n steps. Can you rotate the vector in time proportional to n using only a few dozen extra bytes of storage?
4. Given a dictionary of English words, find sets of anagrams. For instance, "pots", "stop", and "tops" are all anagrams of one another because each can be found by permuting the letters of the others.

5. Write functions for the following date problems: given two dates, compute the number of days between them; given a date, return it's day of the week; given a month and year, produce a calendar of the month as an array of characters
6. Given a very long sequence (say, billions or trillions) of bytes, how would you efficiently count the total number of one bits? (i.e. how many bits are turned on in the entire sequence)
7. Although Quicksort uses only O(logn) stack space on the average, it can use linear space in the worst case. Explain why, then modify the program to use only logarithmic space in the worst case.
8. Write a program for finding the kth-smallest element in the array x[0...n-1] in O(n) expected time. Your algorithm may permute the elements of x.
9. Build the fastest possible complete function to generate a sorted array of random integers without duplicates. (You need feel constrained to use any prescribed interface for set representation)
10. Implement heap-based priority queues to run as quickly as possible; at what values of n are they faster than sequential structures

Other useful Links:

> Java standards
> Google's Code Style Guide
> Google's Android Style Guide

Product links:
Doubleclick
Doubleclick for Advertisers

AdTech Products the team covers:

- **YouTube**: Advertisers' engagement with YouTube can include homepage, masthead, complex, and customized campaigns of over $100,000 dollars / day.
- **DoubleClick Rich Media** (site): This suite of products includes solutions that encompass the most creative and engaging digital ad formats available, like rich media, in-stream, mobile rich media, and dynamic ads. The team also provides the tools and people that support these solutions.
- **Google AdWords**: An application allowing Advertisers to run ads on the web. AdWords ads can be shown on the right side or the top of the Google results page, on syndicated partner sites, and on any site on the web that either uses AdSense to supply ads to their site or is part of the Google Display Network (GDN). AdWords ads are served to these sites via content targeting. Currently, AdWords is sold on a cost-per-click basis, with prices chosen by an auction.
- **Google Display Network**: A network for display ads that includes a collection of websites (display partners), YouTube, and specific Google properties that display AdWords ads. Ads on the GDN can be in text, image, video, or rich media format.
- **AdMob**: A mobile advertising company that Google acquired in 2010, this Google product provides app developers tools to promote, measure, and monetize mobile apps.
- **Dynamic Remarketing**: This "product" refers to a complex ad format that often spans multiple sites. For example, a user might show interest in a pair of boots on Zappos and then see an ad for that same pair of boots elsewhere online. Media Services implements and optimizes these advanced ads.
- **DoubleClick for Advertisers**: DoubleClick for Advertisers (Formerly known as DART For Advertisers) is an ad management and ad serving solution that helps agencies and advertisers manage the entire scope of digital advertising programs.
- **DoubleClick Bid Manager**: DoubleClick Bid Manager is Google's cross-exchange real-time-bidding buying solution for large customers.

- **DoubleClick Search**: DoubleClick Search is a 3rd party tool for managing search campaigns on Google AdWords, Yahoo! Sponsored Search, and MSN adCenter. DoubleClick Search allows agencies and advertisers to manage large numbers of Campaigns, Ad Groups, Keywords, and Ads on multiple search engines, in one interface.Â
- **DoubleClick Ad Exchange**: DoubleClick Ad Exchange is a real-time marketplace that helps ad networks, agency holding companies, and emerging third-party technology providers maximize ROI across millions of sites.

Other Products covered:

- Wallet
- Android Pay
- Hotel Ads
- Transit
- Google Now
- Google Play Books
- Google Play Newsstand
- Crisis Response
- Elections
- Product Reviews / Seller ratings
- App indexing
- Knowledge data for Google Search
- Forms of payments for Google Play and Adwords/Doubleclick

Other useful links:

http://www.thinkwithgoogle.com/insights/
Hangout On Air: Reimagining Technical Support
How Google Hires
Google Philosophy